

习题 9.5

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2025 年 12 月 7 日

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f 为连续函数，由微积分学基本定理可知，可设对任意一点 a ，我们有

$$F(x) = \int_a^x f(t)dt$$

于是

$$\begin{aligned} \frac{d}{dx} \int_{u(x)}^{v(x)} f(t)dt &= \frac{d}{dx} \left(\int_{u(x)}^a f(t)dt + \frac{d}{dx} \int_a^{v(x)} f(t)dt \right) \\ &= \frac{d}{dx} \left(- \int_a^{u(x)} f(t)dt + \frac{d}{dx} \int_a^{v(x)} f(t)dt \right) \\ &= \frac{d}{dx} (-F(u(x)) + F(v(x))) \\ &= -F'(u(x))u'(x) + F'(v(x))v'(x) \end{aligned}$$

严格的来说 $\int_a^{u(x)} f(t)dt = F(u(x))$ 可能需要证明，可以利用定理 9.9，定理 9.10 的证明过程，来证明把积分上限改为 $u(x)$ 也是成立的，过程比较多，这里省略。

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$$\begin{aligned} F(x) &= \int_a^x f(t)(x-t)dt \\ &= \int_a^x f(t)x - f(t)t dt \\ &= \int_a^x f(t)x dt - \int_a^x f(t)t dt \\ &= x \int_a^x f(t)dt - \int_a^x f(t)t dt \end{aligned}$$

利用微积分学基本定理

$$\begin{aligned} F'(x) &= \int_a^x f(t)dt + xf(x) - f(x)x \\ &= \int_a^x f(t)dt \end{aligned}$$

于是

$$F''(x) = f(x)$$

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- (1)

– 方法一：利用微积分学基本定理

因为 $\cos t^2$ 函数在 $\mathbb{R}$ 上连续，利用积分第一中值定理，存在 $\xi \in (0, x)$ ，使得

$$\int_0^x \cos t^2 dt = \cos \xi^2 (x - 0) = x \cos \xi^2$$

所以

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1}{x} \int_0^x \cos t^2 dt &= \lim_{x \rightarrow 0} \frac{1}{x} x \cos \xi^2 \\ &= \lim_{x \rightarrow 0} \cos \xi^2 \end{aligned}$$

因为 $x \rightarrow 0$, $\xi \rightarrow 0$, 所以

$$\begin{aligned}\lim_{x \rightarrow 0} \cos \xi^2 &= \cos 0 \\ &= 1\end{aligned}$$

– 方法二：利用微积分学基本定理

因为 $\cos t^2$ 函数在 $\mathbb{R}$ 上连续, 由微积分学基本定理, $\cos t^2$ 有原函数, 设为 $F(t)$, 又由定理 9.9 可知, $F(x)$ 是连续的, $\lim_{x \rightarrow 0} F(x) = F(0)$, 于是利用洛必达定义

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{1}{x} \int_0^x \cos t^2 dt &= \lim_{x \rightarrow 0} \frac{F(x) - F(0)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\cos x^2}{1} \\ &= 1\end{aligned}$$

• (2)

因为 $e^x \geq 1, x \in [0, +\infty)$, 由积分不等式可知, 分式是未定式 $\frac{\infty}{\infty}$, 使用洛必达法则

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\left(\int_0^x e^{t^2} dt\right)^2}{\int_0^x e^{2t^2} dt} &= \lim_{x \rightarrow 0} \frac{2e^{x^2} \int_0^x e^{t^2} dt}{e^{2x^2}} \\ &= \lim_{x \rightarrow 0} \frac{2 \int_0^x e^{t^2} dt}{e^{x^2}} \\ &= \lim_{x \rightarrow 0} \frac{2e^{x^2}}{2xe^{x^2}} \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \\ &= 0\end{aligned}$$

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• (2)

提示: $x = 2\sin x$.

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- (1)

$$\int_{-a}^a f(x)dx = \int_{-a}^0 f(x)dx + \int_0^a f(x)dx$$

对第一个定积分使用换元积分法, 令 $x = -t, dx = -dt$, 当 t 由 a 变为 0 时, x 由 $-a$ 增到 0 , 于是

$$\begin{aligned}\int_{-a}^0 f(x)dx &= -\int_a^0 f(-t)dt \\ &= \int_0^a f(-t)dt \\ &= \int_0^a f(-x)dx\end{aligned}$$

由奇函数可知 $f(x) + f(-x) = 0$, 所以

$$\begin{aligned}\int_{-a}^a f(x)dx &= \int_{-a}^0 f(x)dx + \int_0^a f(x)dx \\ &= \int_0^a f(-x)dx + \int_0^a f(x)dx \\ &= \int_0^a f(-x) + f(x)dx \\ &= \int_0^a 0dx \\ &= 0\end{aligned}$$

- (2)

证明与 (1) 类似。

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$$\begin{aligned}\int_a^{a+p} f(x)dx &= \int_a^0 f(x)dx + \int_0^p f(x)dx + \int_p^{a+p} f(x)dx \\ &= -\int_0^a f(x)dx + \int_0^p f(x)dx + \int_p^{a+p} f(x)dx\end{aligned}$$

令 $x = t + p, dx = dt$, 当 t 由 0 变为 a 时, x 由 p 增到 $a + p$, 于是

$$\begin{aligned}\int_p^{a+p} f(x)dx &= \int_0^a f(t+p)dt \\ &= \int_0^a f(t)dt \\ &= \int_0^a f(x)dx\end{aligned}$$

所以

$$\begin{aligned}\int_a^{a+p} f(x)dx &= -\int_0^a f(x)dx + \int_0^p f(x)dx + \int_p^{a+p} f(x)dx \\ &= -\int_0^a f(x)dx + \int_0^p f(x)dx + \int_0^a f(x)dx \\ &= \int_0^p f(x)dx\end{aligned}$$

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• (1)

令 $x = \frac{\pi}{2} - t, dx = -dt$, 当 t 由 $\frac{\pi}{2}$ 变为 0 时, x 由 0 增到 $\frac{\pi}{2}$, 于是

$$\begin{aligned}\int_0^{\frac{\pi}{2}} f(\sin x)dx &= -\int_{\frac{\pi}{2}}^0 f(\sin(\frac{\pi}{2} - t))dt \\ &= \int_0^{\frac{\pi}{2}} f(\cos t)dt \\ &= \int_0^{\frac{\pi}{2}} f(\cos x)dx\end{aligned}$$

• (2)

令 $x = \pi - t, dx = -dt$, 当 t 由 π 变为 0 时, x 由 0 增到 π , 于是

$$\begin{aligned}\int_0^{\pi} xf(\sin x)dx &= -\int_{\pi}^0 (\pi - t)f(\sin(\pi - t))dt \\ &= \int_0^{\pi} (\pi - t)f(\sin t)dt \\ &= \pi \int_0^{\pi} f(\sin t)dt - \int_0^{\pi} tf(\sin t)dt \\ &= \pi \int_0^{\pi} f(\sin x)dx - \int_0^{\pi} xf(\sin x)dx\end{aligned}$$

所以

$$\begin{aligned} 2 \int_0^\pi x f(\sin x) dx &= \int_0^\pi f(\sin x) dx \\ \int_0^\pi x f(\sin x) dx &= \frac{\pi}{2} \int_0^\pi f(\sin x) dx \end{aligned}$$

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(i)

$$\begin{aligned} J(m, n) &= \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx \\ &= \frac{1}{m+1} \int_0^{\frac{\pi}{2}} \cos^{n-1} x d(\sin^{m+1} x) \\ &= \frac{1}{m+1} \left(\cos^{n-1} x \sin^{m+1} x \Big|_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{m+2} x \cos^{n-2} x dx \right) \\ &= \frac{n-1}{m+1} \int_0^{\frac{\pi}{2}} \sin^{m+2} x \cos^{n-2} x dx \\ &= \frac{n-1}{m+1} \int_0^{\frac{\pi}{2}} \sin^m x (1 - \cos^2 x) \cos^{n-2} x dx \\ &= \frac{n-1}{m+1} \left(\int_0^{\frac{\pi}{2}} \sin^m x \cos^{n-2} x dx - \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx \right) \\ &= \frac{n-1}{m+1} (J(m, n-2) - J(m, n)) \end{aligned}$$

于是可得

$$\begin{aligned} J(m, n) + \frac{n-1}{m+1} J(m, n) &= \frac{n-1}{m+1} J(m, n-2) \\ \frac{m+n}{m+1} J(m, n) &= \frac{n-1}{m+1} J(m, n-2) \\ J(m, n) &= \frac{n-1}{m+n} J(m, n-2) \end{aligned}$$

令 $t = \frac{\pi}{2} - x, dx = -dt$, 于是

$$\begin{aligned}
J(m, n) &= \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx \\
&= - \int_{\frac{\pi}{2}}^0 \sin^m \left(\frac{\pi}{2} - t \right) \cos^n \left(\frac{\pi}{2} - t \right) dt \\
&= \int_0^{\frac{\pi}{2}} \sin^m \left(\frac{\pi}{2} - t \right) \cos^n \left(\frac{\pi}{2} - t \right) dt \\
&= \int_0^{\frac{\pi}{2}} \cos^m t \sin^n t dt \\
&= J(n, m)
\end{aligned}$$

利用以上结论, 我们有

$$\begin{aligned}
J(m, n) &= J(n, m) \\
&= \frac{m-1}{n+m} J(n, m-2) \\
&= \frac{m-1}{m+n} J(m-2, n)
\end{aligned}$$

(ii)

$$\begin{aligned}
J(2m, 2n) &= \frac{2m-1}{2m+2n} J(2m-2, 2n) \\
&= \frac{2m-1}{2m+2n} \cdot \frac{2m-3}{2m+2n-2} J(2m-4, 2n) \\
&= \frac{2m-1}{2m+2n} \cdot \frac{2m-3}{2m+2n-2} \cdots \frac{1}{2n+2} J(0, 2n) \\
&= \frac{(2m-1)!!}{(2m+2n)(2m+2n-2) \cdots (2n+2)} J(0, 2n)
\end{aligned}$$

另外

$$\begin{aligned}
J(0, 2n) &= \frac{2n-1}{2n} J(0, 2n-2) \\
&= \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} J(0, 2n-4) \\
&= \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdots \frac{1}{2} J(0, 0)
\end{aligned}$$

于是可得

$$\begin{aligned}
 J(2m, 2n) &= \frac{(2m-1)!!}{(2m+2n)(2m+2n-2) \cdot (2n+2)} J(0, 2n) \\
 &= \frac{(2m-1)!!}{(2m+2n)(2m+2n-2) \cdot (2n+2)} \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdots \frac{1}{2} J(0, 0) \\
 &= \frac{(2m-1)!!(2n-1)!!}{(2m+2n)!!} J(0, 0) \\
 &= \frac{(2m-1)!!(2n-1)!!}{(2m+2n)!!} \cdot \frac{\pi}{2}
 \end{aligned}$$

注：为了计算方便列出一下定积分的值

$$\begin{aligned}
 J(0, 0) &= \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{2} \\
 J(1, 0) &= \int_0^{\frac{\pi}{2}} \sin x dx = 1 \\
 J(0, 1) &= \int_0^{\frac{\pi}{2}} \cos x dx = 1 \\
 J(1, 1) &= \int_0^{\frac{\pi}{2}} \sin x \cos x dx = \frac{1}{2}
 \end{aligned}$$

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由习题 1 可知

$$g'(x) = af(ax) - f(x) = 0$$

即

$$f(x) = af(ax)$$

对任意 $x > 0$, 取 $a = \frac{1}{x}$, 有

$$\begin{aligned}
 f(x) &= \frac{1}{x} f\left(\frac{1}{x}x\right) \\
 &= \frac{1}{x} f(1)
 \end{aligned}$$

令 $c = f(1)$, $f(x) = \frac{c}{x}$, 命题成立。

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注意：题中的“连续可微函数”，指的是：导函数是连续的。

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设两个阴影从左到右分别为 S_1, S_2 , 于是可得

$$\begin{aligned} S_1 &= \int_a^\xi f(x)dx - \int_a^\xi f(a)dx = \int_a^\xi f(x) - f(a)dx \\ S_2 &= \int_\xi^b f(b)dx - \int_\xi^b f(x)dx = \int_\xi^b f(b) - f(x)dx \end{aligned}$$

于是

$$S_1 - S_2 = \int_a^\xi f(x) - f(a)dx - \int_\xi^b f(b) - f(x)dx$$

定义辅助函数

$$F(x) = \int_a^x f(t) - f(a)dt - \int_x^b f(b) - f(t)dt$$

由题设和微积分基本定理可知, $F(x)$ 是连续可导函数, 且

$$\begin{aligned} F(a) &= \int_a^a f(t) - f(a)dt - \int_a^b f(b) - f(t)dt \\ &= - \int_a^b f(b) - f(t)dt \end{aligned}$$

由与 $f(x)$ 是 $[a, b]$ 上的严格增的连续函数, 所以 $f(b) - f(t) > 0, t \in [a, b]$, 可得

$$F(a) = - \int_a^b f(b) - f(t)dt < 0$$

同理

$$\begin{aligned} F(b) &= \int_a^x f(t) - f(a)dt - \int_x^b f(b) - f(t)dt \\ &= \int_a^b f(t) - f(a)dt - \int_b^b f(b) - f(t)dt \\ &= \int_a^b f(t) - f(a)dt \\ &> 0 \end{aligned}$$

由连续函数的介质性定理可知, 存在 $\xi \in (a, b)$, 使得

$$\int_a^\xi f(x) - f(a)dx - \int_\xi^b f(b) - f(x)dx = 0$$

即

$$S_1 - S_2 = 0$$

命题得证。

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- 方法一:

利用第二积分中值定理的推论, 存在 $\xi \in [0, 2\pi]$, 使得

$$\begin{aligned}\int_0^{2\pi} f(x) \sin nx dx &= f(0) \int_0^\xi \sin nx dx + f(2\pi) \int_\xi^{2\pi} \sin nx dx \\&= f(0) \left(-\frac{1}{n} \cos nx \Big|_0^\xi \right) + f(2\pi) \left(-\frac{1}{n} \cos nx \Big|_\xi^{2\pi} \right) \\&= f(0) \left[-\frac{1}{n} (\cos n\xi - 1) \right] + f(2\pi) \left[-\frac{1}{n} (1 - \cos n\xi) \right] \\&= (1 - \cos n\xi) \frac{f(0) - f(2\pi)}{n} \\&\geq 0\end{aligned}$$

- 方法二:

令 $h(x) = f(x) - f(2\pi)$, 因为 $f(x)$ 为单调递减函数, 则 h 为非负、递减函数。由定理 9.11(i), 存在 $\xi \in [0, 2\pi]$, 使得

$$\begin{aligned}\int_0^{2\pi} h(x) \sin nx dx &= \int_0^{2\pi} (f(x) - f(2\pi)) \sin nx dx \\&= h(0) \int_0^\xi \sin nx dx\end{aligned}$$

因为

$$\begin{aligned}
\int_0^{2\pi} f(x) \sin nx dx &= \int_0^{2\pi} (f(x) - f(2\pi)) \sin nx + f(2\pi) \sin nx dx \\
&= \int_0^{2\pi} h(x) \sin nx dx + f(2\pi) \int_0^{2\pi} \sin nx dx \\
&= h(0) \int_0^\xi \sin nx dx + 0 \\
&= h(0) \left(-\frac{1}{n} \cos nx \right) \Big|_0^\xi \\
&= h(0) \left(-\frac{1}{n} (\cos n\xi - 1) \right) \\
&= h(0) \frac{1 - \cos n\xi}{n} \\
&\geq 0
\end{aligned}$$

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令 $t^2 = u, t = \sqrt{u}, dt = \frac{1}{2\sqrt{u}} du$, 于是

$$\int_x^{x+c} \sin t^2 dt = \int_{x^2}^{(x+c)^2} \frac{\sin u}{2\sqrt{u}} du$$

由于 $\frac{1}{2\sqrt{u}}$ 在 $[x^2, (x+c)^2]$ 是单调递减的, 且 $\frac{1}{2\sqrt{u}} > 0$, 由积分第二中值定理可知, 存在 $\xi \in [x^2, (x+c)^2]$, 使得

$$\begin{aligned}
\int_{x^2}^{(x+c)^2} \frac{\sin u}{2\sqrt{u}} du &= \frac{1}{2\sqrt{x^2}} \int_{x^2}^\xi \sin u du \\
&= \frac{1}{2x} (-\cos u) \Big|_{x^2}^\xi \\
&= \frac{1}{2x} (\cos x^2 - \cos \xi)
\end{aligned}$$

因为

$$-2 \leq |\cos x^2 - \cos \xi| \leq 2$$

所以

$$-\frac{1}{x} \leq \left| \frac{1}{2x} (\cos x^2 - \cos \xi) \right| \leq \frac{1}{x}$$

即

$$\left| \int_x^{x+c} \sin t^2 dt \right| \leq \frac{1}{x}$$

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todo

首先要确定 $f(\varphi(t))\varphi'(t)$ 是可积的。

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令

$$g'(x) = \begin{cases} f'(x), & f'(x) \geq 0 \\ 0, & f'(x) < 0 \end{cases}, h'(x) = \begin{cases} f'(x), & f'(x) \leq 0 \\ 0, & f'(x) > 0 \end{cases},$$

因为 f 在 $[a, b]$ 上是连续可微的, 所以 $g'(x), h'(x)$ 都是连续, 令 $g(x), h(x)$ 分别是 $\int_a^x g'(t)dt, \int_a^x h'(t)dt$ 一个原函数, 于是

$$\begin{aligned} \int_a^x f'(t)dt &= \int_a^x g'(t) + h'(t)dt \\ f(x) - f(a) &= (g(t) + h(t)) \Big|_a^x \\ &= g(x) + h(x) - g(a) - h(a) \end{aligned}$$

可得

$$f(x) = g(x) + h(x) + f(a) - g(a) - h(a)$$

取 $C = f(a) - g(a) - h(a)$ 。综上, $g(x)$ 是增函数, $h(x) = h(x) + C$ 任然是减函数。